Supplementary Experimental Results for VERMITHOR:

Formally Verified Runtime Orchestration
for Thermally Safe Edge CPS Inference

This document provides the extended experimental results referenced in the main paper. All experiments are conducted on the NVIDIA Jetson Thor with TensorRT FP16 inference, Mesh-Exit ResNet-18 backbone, and the evaluation setup described in Section V-A of the main paper. Unless otherwise noted, results are reported as mean ± std over 5 random seeds (42–46).

Contents

1 Per-Scenario Safety Violation Breakdowns (RQ1) 2
- 1.1 Thermal Violation Rates (%) . . . 2
- 1.2 Network Violation Rates (%) . . . 2

2 Cross-Dataset Results 3
- 2.1 Cross-Dataset Safety Violations (RQ1) . . . 3
- 2.2 Cross-Dataset Hardware Efficiency (RQ2) . . . 3
- 2.3 Cross-Dataset CP Coverage (RQ3) . . . 4

3 Mesh-Exit DAG Architecture Details 4
- 3.1 Per-Exit Accuracy and Latency . . . 4
- 3.2 Bottleneck Encoder Dimensions . . . 4

4 Extended Ablation Study (RQ5) 5
- 4.1 Pairwise Component Removal . . . 5
- 4.2 Dwell-Time Sensitivity ($\tau_{\min}$) . . . 5

5 Hyperparameter Sensitivity 6
- 5.1 Disentangled Loss Weights (α, β, γ) . . . 6
- 5.2 RCP Inflation Sensitivity (β_{inf}) . . . 6
- 5.3 PID Controller Parameters . . . 7

6 Computational Overhead Breakdown 7

7 Mode Distribution Analysis 7

1 Per-Scenario Safety Violation Breakdowns (RQ1)

The main paper reports violation rates across five consolidated evaluation conditions. Tables 1–2 provide the complete per-scenario breakdowns across all seven stress scenarios for all baselines.

1.1 Thermal Violation Rates (%)

Table 1: Per-scenario thermal violation rate (%, mean $\pm$ std, 5 seeds, CIFAR-100). Violations are defined as $T_t > T_{\text{lim}} = 85\,^{\circ}\text{C}$.

Method	**Normal**	**High Temp**	**Net. Deg.**	**Tun. (S)**	**Tun. (F)**	**Osc. BW**	**Combined**
VERMITHOR (ours)	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0
BranchyNet	0.0±0.0	12.7±3.8	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	18.4±5.2
Neurosurgeon	0.0±0.0	14.1±4.3	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	17.8±4.9
MDI-Exit	0.0±0.0	11.9±3.5	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	16.2±4.7
SAGE	0.0±0.0	13.4±4.1	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	19.1±5.6
ThermalGuard	0.0±0.0	2.1±1.2	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	5.7±2.3
No Monitor	0.0±0.0	15.6±4.8	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	21.3±6.1
Oracle	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0

Observations. Thermal violations concentrate exclusively in the High Temp and Combined Stress scenarios. All non-thermal-aware baselines (BranchyNet, Neurosurgeon, MDI-Exit, SAGE, No Monitor) exhibit structurally identical failure patterns: zero violations under normal/network-only conditions, 11.9–15.6% under High Temp, and 16.2–21.3% under Combined Stress. ThermalGuard reduces thermal violations to 2.1% and 5.7% respectively, but cannot eliminate them entirely because its reactive threshold policy (switch at $T > 80\,^{\circ}\text{C}$) lacks the predictive safe-horizon mechanism of VERMITHOR's Lipschitz bound (Theorem 1 in the main paper).

1.2 Network Violation Rates (%)

Table 2: Per-scenario network violation rate (%, mean $\pm$ std, 5 seeds, CIFAR-100). Violations are defined as offloading when $\text{BW} < \text{BW}_{\text{min}} = 5\,\text{Mbps}$ or $\text{RTT} > \text{RTT}_{\text{max}} = 100\,\text{ms}$.

Method	**Normal**	**High Temp**	**Net. Deg.**	**Tun. (S)**	**Tun. (F)**	**Osc. BW**	**Combined**
VERMITHOR (ours)	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0
BranchyNet	0.0±0.0	0.0±0.0	4.3±1.8	8.7±2.4	14.2±3.1	2.1±0.9	11.6±3.7
Neurosurgeon	0.0±0.0	0.0±0.0	18.7±4.2	29.1±5.3	44.2±6.8	9.4±2.8	31.8±5.9
MDI-Exit	0.0±0.0	0.0±0.0	7.8±2.3	12.4±3.1	21.6±4.5	3.7±1.4	16.9±4.2
SAGE	0.0±0.0	0.0±0.0	5.6±1.9	10.3±2.7	17.8±3.8	2.9±1.1	14.1±3.9
ThermalGuard	0.0±0.0	0.0±0.0	11.2±3.1	18.4±4.0	26.3±5.1	5.8±2.0	22.7±4.8
No Monitor	0.0±0.0	0.0±0.0	22.1±5.1	34.7±6.2	58.4±7.9	12.8±3.4	41.3±7.1
Oracle	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0

Observations. Network violations scale with scenario severity: Tunnel (Full) produces the highest violation rates across all baselines (14.2–58.4%), followed by Combined Stress (11.6–41.3%), and Tunnel (Short) (8.7–34.7%). Neurosurgeon exhibits the worst network safety because its static latency-minimization objective concludes that local inference is always faster on Jetson Thor, effectively disabling offload—yet when thermal pressure forces offload consideration, it lacks any mechanism to detect unsafe network conditions. ThermalGuard's network violations (11.2–26.3%)

are higher than BranchyNet, MDI-Exit, and SAGE because its single-dimensional thermal focus leads it to offload aggressively under heat, ignoring network degradation.

2 Cross-Dataset Results

The main paper reports CIFAR-100 as the primary evaluation dataset. Tables 3–4 confirm that all findings hold across CIFAR-10 and ImageNet-120 (120 classes, 224×224 resolution, custom subset of ImageNet-1K).

2.1 Cross-Dataset Safety Violations (RQ1)

Table 3: Safety violation rate (%) under Combined Stress, three datasets (mean $\pm$ std, 5 seeds).

	CIFAR-10		**CIFAR-100**		**ImageNet-120**	
Method	Therm.	Net.	Therm.	Net.	Therm.	Net.
VERMITHOR (ours)	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0
BranchyNet	17.8±5.0	10.9±3.5	18.4±5.2	11.6±3.7	19.2±5.4	12.3±3.9
Neurosurgeon	17.2±4.7	30.4±5.7	17.8±4.9	31.8±5.9	18.5±5.1	33.1±6.2
MDI-Exit	15.8±4.5	16.1±4.0	16.2±4.7	16.9±4.2	16.9±4.8	17.6±4.4
SAGE	18.6±5.4	13.5±3.7	19.1±5.6	14.1±3.9	19.8±5.8	14.8±4.1
ThermalGuard	5.4±2.1	21.9±4.6	5.7±2.3	22.7±4.8	6.1±2.4	23.4±5.0
No Monitor	20.8±5.9	39.7±6.9	21.3±6.1	41.3±7.1	22.1±6.3	42.8±7.4
Oracle	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0

Observations. Violation rates are consistent across datasets, with ImageNet-120 showing marginally higher rates (+0.5–1.5 pp) due to the heavier computational load of 224×224 inputs increasing thermal pressure and offload frequency. VERMITHOR and Oracle maintain 0% violations universally.

2.2 Cross-Dataset Hardware Efficiency (RQ2)

Table 4: Hardware efficiency under Combined Stress, three datasets (mean $\pm$ std, 5 seeds, Jetson Thor).

	Mean Latency (ms)↓			**Energy (mJ/img)↓**		
Method	C-10	C-100	IN-120	C-10	C-100	IN-120
VERMITHOR (ours)	1.38±0.12	1.42±0.14	3.87±0.31	34.6±3.1	35.5±3.3	96.8±7.8
BranchyNet	1.14±0.09	1.18±0.11	3.21±0.24	28.5±2.4	29.5±2.6	80.3±6.1
Neurosurgeon	2.41±0.08	2.49±0.09	6.78±0.41	60.3±2.1	62.3±2.3	169.5±10.3
MDI-Exit	1.78±0.16	1.84±0.18	5.01±0.45	44.5±4.1	46.0±4.3	125.3±11.3
SAGE	1.89±0.14	1.95±0.16	5.31±0.42	47.3±3.6	48.8±3.8	132.8±10.5
ThermalGuard	1.51±0.11	1.56±0.13	4.25±0.33	37.8±3.0	39.0±3.2	106.3±8.3
No Monitor	2.41±0.08	2.49±0.09	6.78±0.41	60.3±2.1	62.3±2.3	169.5±10.3
Oracle	1.29±0.10	1.33±0.12	3.62±0.28	32.3±2.7	33.3±2.9	90.5±7.1

Observations. The relative efficiency ordering is preserved across all three datasets. ImageNet-120 latencies are approximately 2.7× higher than CIFAR due to the 7× larger input resolution

(224^2 vs. 32^2), but the VERMITHOR-to-Oracle gap remains stable at 7–8%. CIFAR-10 and CIFAR-100 latencies are nearly identical, confirming that classification head size (10 vs. 100 outputs) has negligible impact on inference time.

2.3 Cross-Dataset CP Coverage (RQ3)

Table 5: Empirical CP coverage (%) at shift intensity $\gamma = 1.0$ (worst case), three datasets. Target: 90%.

Method	**CIFAR-10**	**CIFAR-100**	**ImageNet-120**
VERMITHOR-RCP	89.6±0.7	89.4±0.8	89.1±0.9
Vanilla CP	56.2±3.4	55.7±3.6	54.8±3.9
Weighted CP	53.4±3.9	52.9±4.1	51.7±4.4
MC-Dropout	47.8±5.9	47.3±6.2	46.1±6.7
No-CP	40.7±6.8	40.1±7.1	38.9±7.5

Observations. VERMITHOR-RCP maintains ≥89.1% coverage across all datasets, within 0.9 pp of the 90% target even at maximum shift. All non-adaptive methods collapse below 57% at $\gamma = 1.0$. ImageNet-120 shows marginally lower coverage (−0.5 pp vs. CIFAR-100) because higher-dimensional features increase the effective f-divergence between calibration and runtime distributions.

3 Mesh-Exit DAG Architecture Details

3.1 Per-Exit Accuracy and Latency

Table 6 reports the accuracy and inference latency of each exit point in the Mesh-Exit ResNet-18 architecture, demonstrating the accuracy–latency trade-off exploited by the orchestrator.

Table 6: Per-exit performance on CIFAR-100 (Jetson Thor, TensorRT FP16, batch = 1, 1000 runs).

Exit Point	**Depth**	**Acc. (%)**	**Lat. (ms)**	**Energy (mJ)**
Exit 0 (earliest)	After Block 1	58.3±0.4	0.64±0.03	16.0±0.8
Exit 1	After Block 2	65.7±0.3	1.12±0.05	28.0±1.3
Exit 2	After Block 3	69.4±0.3	1.78±0.07	44.5±1.8
Full Backbone	After Block 4	70.4±0.2	2.49±0.09	62.3±2.3
Super-Node topology				
SN Exit 0 (local)	Block 1 + SN head	59.9±0.4	0.71±0.04	17.8±0.9
SN Exit 1 (local)	Block 2 + SN head	66.2±0.3	1.19±0.05	29.8±1.4

Observations. Exit 0 achieves 3.9× latency speedup over the full backbone (0.64 vs. 2.49 ms) at a 12.1 pp accuracy cost. The Super-Node heads (SN Exit 0, SN Exit 1) add only 0.07–0.11 ms overhead while providing +1.6 pp and +0.5 pp accuracy improvements over their base exits, confirming the disentangled training objective preserves rather than degrades performance. The orchestrator dynamically selects among these exits based on the current thermal and network state.

3.2 Bottleneck Encoder Dimensions

All bottleneck encoders are single-layer linear projections with ReLU activation, contributing $<$ 0.02 ms additional latency each.

Table 7: Bottleneck encoder configuration for split-offload modes.

Split Point	Feature Dim.	Bottleneck Dim.	Compression
After Block 1	$16 \times 16 \times 64$	512	32.0×
After Block 2	$8 \times 8 \times 128$	256	32.0×
After Block 3	$4 \times 4 \times 256$	128	32.0×

4 Extended Ablation Study (RQ5)

4.1 Pairwise Component Removal

The main paper reports single-component ablation (remove one of STL, RCP, or ME at a time). Table 8 extends this to pairwise removal, confirming that no pair of components compensates for the removal of the third.

Table 8: Pairwise component ablation under Combined Stress (CIFAR-100, 5 seeds). "✓" = present, "—" = removed.

STL	RCP	ME	Viol. (%)	Cov. (%)	Lat. (ms)	Sw./100	P99 (ms)
✓	✓	✓	**0.0±0.0**	**90.1±0.5**	**1.42±0.14**	**4.2±0.6**	**3.87**
✓	✓	—	0.0±0.0	89.8±0.6	2.49±0.09	3.8±0.5	5.12
✓	—	✓	0.0±0.0	79.3±1.8	1.47±0.15	4.5±0.7	4.21
—	✓	✓	8.4±2.1	90.0±0.5	1.39±0.13	19.4±2.7	11.2
✓	—	—	0.0±0.0	78.9±1.9	2.49±0.09	3.5±0.4	5.43
—	✓	—	11.2±2.8	89.7±0.6	2.49±0.09	0.0	5.12
—	—	✓	14.7±3.4	76.4±2.3	1.44±0.14	22.8±3.1	13.8
—	—	—	21.3±6.1	68.3±3.5	2.49±0.09	0.0	5.12

Observations.

- **STL is necessary for safety**: Removing STL (rows 4, 6, 7, 8) introduces 8.4–21.3% violations regardless of which other components are present. RCP+ME alone (row 4) still violates because the prediction region informs but does not *enforce* safety.
- **RCP is necessary for coverage**: Removing RCP (rows 3, 5, 7, 8) drops coverage from 90.1% to 76.4–79.3%. STL+ME alone (row 3) maintains safety but with 10.8 pp coverage loss.
- **ME is necessary for efficiency**: Removing ME (rows 2, 5, 6, 8) forces the full backbone (2.49 ms), eliminating the 1.75× latency advantage. STL+RCP alone (row 2) preserves safety and coverage but at +75% latency cost.
- **No pair substitutes for the third**: The minimum violation rate without STL is 8.4%; the minimum coverage without RCP is 79.3%; the minimum latency without ME is 2.49 ms. These represent distinct, non-overlapping failure modes aligned with the three dimensions of the Edge Trilemma.

4.2 Dwell-Time Sensitivity ($\tau_{\min}$)

Observations. $\tau_{\min} = 20$ represents the Pareto-optimal operating point: ping-pong oscillations are eliminated ($\leq 0\%$) while maintaining zero safety violations. At $\tau_{\min} = 50$, the excessively long dwell window introduces rare thermal violations (0.8%) because the controller cannot react to rapid thermal transients within the locked dwell period. At $\tau_{\min} = 5$, mode oscillation resurfaces (8.7% ping-pong), defeating the purpose of the dwell constraint.

Table 9: Effect of dwell-time parameter $\tau_{\min}$ on mode stability and latency (Combined Stress, CIFAR-100, 5 seeds).

$\tau_{\min}$ (steps)	Sw./100	Dwell (s)	Ping-Pong (%)	P99 Lat. (ms)	Viol. (%)
5	11.8±1.8	3.2±0.5	8.7±1.9	6.24	0.0±0.0
10	7.4±1.1	5.6±0.9	2.3±0.8	4.81	0.0±0.0
20	**4.2±0.6**	**8.7±1.2**	**0.0**	**3.87**	**0.0±0.0**
30	3.1±0.4	11.8±1.6	0.0	3.72	0.0±0.0
50	2.4±0.3	16.3±2.1	0.0	3.64	0.8±0.4

5 Hyperparameter Sensitivity

5.1 Disentangled Loss Weights (α, β, γ)

The training loss for the Mesh-Exit DAG is $\mathcal{L} = \alpha\mathcal{L}_{\text{cls}} + \beta\mathcal{L}_{\text{exit}} + \gamma\mathcal{L}_{\text{btn}}$, with default weights $\alpha = 0.3$, $\beta = 0.5$, $\gamma = 0.2$. Table 10 reports the sensitivity analysis.

Table 10: Effect of loss weight configurations on backbone accuracy and exit quality (CIFAR-100, 5 seeds).

α	β	γ	Full Acc. (%)	Exit 0 Acc. (%)	Exit 1 Acc. (%)	Btn. Recon. (dB)
0.5	0.3	0.2	70.6±0.3	56.1±0.5	63.4±0.4	28.7±0.8
0.3	0.3	0.4	70.2±0.3	55.8±0.5	63.1±0.4	31.4±0.7
0.3	**0.5**	**0.2**	**70.4±0.2**	**58.3±0.4**	**65.7±0.3**	**29.3±0.8**
0.2	0.5	0.3	69.8±0.3	58.1±0.5	65.4±0.4	30.1±0.7
0.2	0.6	0.2	69.4±0.4	58.9±0.5	66.1±0.4	28.9±0.9
0.4	0.4	0.2	70.7±0.2	57.2±0.4	64.6±0.3	29.1±0.8

Observations. Increasing β (exit weight) from 0.3 to 0.5 improves early-exit accuracy by +2.2 pp (Exit 0) and +2.3 pp (Exit 1) with only −0.2 pp backbone degradation, confirming the disentangled objective's effectiveness. Increasing γ (bottleneck weight) improves reconstruction quality (+2.7 dB) at the cost of early-exit accuracy (−0.2 pp), reflecting the trade-off between split-offload fidelity and local exit quality. The default $(\alpha, \beta, \gamma) = (0.3, 0.5, 0.2)$ balances all three objectives.

5.2 RCP Inflation Sensitivity (β_{inf})

Table 11: Effect of RCP inflation sensitivity β_{inf} on coverage and interval width (Combined Stress, $\gamma = 0.6$, CIFAR-100).

β_{inf}	Coverage (%)	Interval Width (ms)	Lat. (ms)	Viol. (%)
0.5	86.3±1.2	11.4	1.38±0.12	0.0
1.0	88.7±0.9	15.8	1.41±0.13	0.0
2.0	**89.9±0.6**	**18.6**	**1.42±0.14**	**0.0**
3.0	90.4±0.5	22.1	1.48±0.16	0.0
5.0	91.2±0.4	29.3	1.62±0.21	0.0

Observations. Increasing β_{inf} monotonically improves coverage but widens the prediction interval, causing the orchestrator to adopt more conservative (higher-latency) modes. At $\beta_{\text{inf}} = 2.0$ (default), coverage reaches 89.9% with modest interval width. At $\beta_{\text{inf}} = 5.0$, coverage exceeds the

target (91.2%) but at +14% latency cost. Safety violations remain zero across all settings because the STL monitor enforces hard constraints independently of the CP coverage level.

5.3 PID Controller Parameters

Table 12: PID controller parameter sensitivity (Combined Stress, CIFAR-100).

Configuration	λ Overshoot (%)	Settling Time (steps)	Coverage (%)	Viol. (%)
β=1.0, κ=0.0 (P-only)	24.3	18	88.4±1.1	0.0
β=2.0, κ=0.0 (PI-only)	14.7	12	89.6±0.7	0.0
β=2.0, κ=0.3 (full PID)	**6.2**	**8**	**89.9±0.6**	**0.0**
β=2.0, κ=1.0 (over-damped)	2.1	22	88.1±0.9	0.0

Observations. The full PID controller ($\kappa = 0.3$) achieves the best balance: 6.2% overshoot with 8-step settling time. The P-only controller oscillates excessively (24.3% overshoot), while the over-damped configuration ($\kappa = 1.0$) settles too slowly (22 steps), during which coverage may be sub-optimal. Safety is unaffected across all PID configurations because the STL monitor provides the hard safety layer independent of the inflation dynamics.

6 Computational Overhead Breakdown

Table 13: Per-step computational overhead breakdown (Jetson Thor, averaged over 10,000 steps).

Component	**Time (μs)**	**% of Control Step**
Monotonic-queue STL evaluation (×3)	12.4±1.8	0.011%
EMA divergence update	1.3±0.2	0.001%
PID inflation control	0.8±0.1	<0.001%
Density-ratio forward pass (r_θ)	47.6±4.2	0.043%
Mode selection (5 modes, 3 specs)	3.1±0.4	0.003%
Total orchestrator overhead	**65.2±5.1**	**0.059%**
For reference:		
Inference (Exit 0, fastest)	640±30	—
Inference (Full backbone)	2,490±90	—
Control step interval (9 Hz)	111,111	—

Observations. The total orchestrator overhead is 65.2 μs per control step, which is <0.06% of the 111 ms control interval and < 10% of the fastest inference path (Exit 0). The density-ratio classifier r_θ dominates (73% of overhead) due to the neural network forward pass; the STL evaluation, EMA, and PID components together consume only 14.5 μs. This confirms Theorem 4 (O(1) amortized complexity) and validates that the formal monitoring machinery imposes negligible computational burden.

7 Mode Distribution Analysis

Observations. The mode distribution reveals the orchestrator's adaptive behavior:

- Under Normal conditions, OFFLOAD dominates (84.2%) because network conditions are favorable and offloading yields the best accuracy.

Table 14: Mode distribution (% of control steps) across evaluation conditions (CIFAR-100, 5 seeds).

Mode	**Normal**	**Thermal**	**Network**	**Combined**	**Adversarial**
m_1: OFFLOAD	84.2±3.1	12.3±4.7	0.0±0.0	8.4±3.2	2.1±1.4
m_2: FULL_LOCAL	8.7±2.4	28.6±5.1	36.4±4.8	22.1±4.3	14.7±3.8
m_3: LOCAL_EXIT	5.3±1.8	52.8±6.2	47.2±5.4	48.7±5.6	41.3±5.1
m_4: SPLIT_OFFLOAD	1.8±0.9	6.1±2.3	14.8±3.7	18.4±4.1	28.6±4.7
m_5: EMERGENCY	0.0±0.0	0.2±0.3	1.6±0.8	2.4±1.1	13.3±3.4

- Under Thermal stress, LOCAL_EXIT becomes dominant (52.8%) as the orchestrator routes through early exits to reduce GPU power draw.
- Under Network stress, the orchestrator shifts entirely away from OFFLOAD (0.0%) to LOCAL_EXIT (47.2%) and FULL_LOCAL (36.4%).
- Under Adversarial conditions, EMERGENCY mode activates on 13.3% of steps—confirming it serves as a genuine fallback rather than the primary operating mode.
- SPLIT_OFFLOAD usage increases under Combined/Adversarial conditions (18.4–28.6%) as the orchestrator leverages bottleneck compression to maintain partial offloading under degraded but non-zero network conditions.